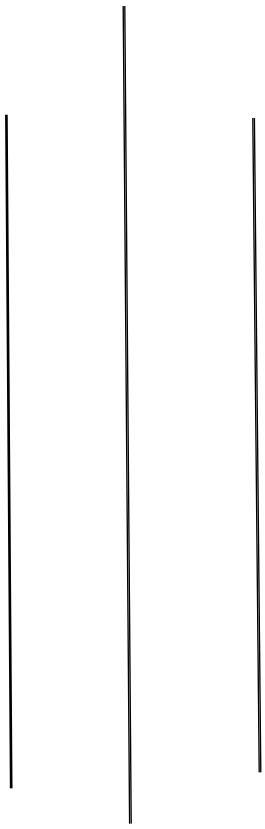
WEEK 1: INTERNSHIP REPORT



Submitted By: Yanji Pakhrin

To

OneCover Nepal

Table of Content

List of Figures.....ii

List of Tables.....iii

CYBERSECURITY, LINUX AND NETWORKING FUNDAMENTALS..... 1

1.1 Introduction..... 1

1.2 Topics Learned..... 1

1.2.1 Introduction to Cybersecurity..... 1

1.2.2 CIA Triad..... 2

1.2.3 OSI Model..... 4

1.2.4 TCP/IP..... 7

1.2.5 IP Address..... 9

1.2.6 TCP VS UDP..... 11

1.2.7 Common ports and Protocols..... 12

1.2.8 Linux File System..... 14

1.2.9 Linux File Permission..... 15

Practical Tasks..... 17

Required Labs..... 26

List of Figures

Figure 1.1: CIA Triad 2
Figure 1.2: OSI Model 4
Figure 1.3: TCP/IP Model 7
Figure 3.1: Task 1 26
Figure 3.2 : Task 2 26
Figure 3.3: Task 3 27
Figure 3.4: Task 4 27
Figure 3.5: Task2 28
Figure 3.6: Task 3 28
Figure 3.7: Task 4 29
Figure 3.8: Task 5 30
Figure 3.9: Task 6 30
Figure 3.10: Task 7 30

List of Tables

Table 1.1: Classes of IPv4……10
Table 1.2: TCP VS UDP……..11
Table 1.3: Well-Known Ports……..12
Table 1.4: Registered Ports……..13
Table 1.5: Specialized and Additional Ports……..14
Table 1.6: Directories……..14
Table 1.7: Basic Linux Commands……..15
Table 1.8: Linux Permission Commands……..16
Table 2.1: Common Ports, Protocol and Purposes……..25

CYBERSECURITY, LINUX AND NETWORKING FUNDAMENTALS

1.1 Introduction

The first week of the cybersecurity internship focused on developing a basic understanding of cybersecurity, networking and Linux fundamentals. The purpose of this week was to build the foundational knowledge required for further learning and practical work in cybersecurity.

During this week, I studied important cybersecurity concepts such as the CIA Triad, as well as networking fundamentals including TCP/IP and OSI models, IP addresses, TCP and UDP, common ports, and network protocols. I also learned about the Linux file system, basic Linux commands, and file permissions.

Along with theoretical learning. I completed practical exercise using Linux and the TryHackMe platforms. I practiced Linux commands, created and manage files and directories, copied and moved files, searched for files, and examined Linux file permissions.

1.2 Topics Learned

1.2.1 Introduction to Cybersecurity

Cybersecurity is the practice of protecting systems, networks, programs, devices, individuals, organizations, data from digital attacks, damage or unauthorized access.

i. Importance of Cybersecurity

- Protects personal information
- Prevents cyberattacks
- Ensures privacy
- Maintains data integrity
- Supports national security

ii. Common cybersecurity threats

- **Virus:** A virus is a type of malware that can copy or reproduce itself by attaching to legitimate files or programs and spread across a system when an infected file is opened.
- **Ransomware:** It is a malware that prevents users from accessing their files or computer systems, usually by encrypting data, and then demands a ransom payment to remote access.
- **Adware:** It is a software that automatically displays unwanted advertisements on user's device, often while browsing the internet or using application.
- **Spyware:** Spyware is a type of malicious software that secretly monitors a user's activities and collects information without the user's knowledge or permissions.
- **Phishing:** Phishing is a cyberattack in which an attacker pretends to be a trusted person or organization to trick users into revealing sensitive information, such as passwords, banking details, or personal data.
- **Trojan Horse:** A Trojan Horse is malware that pretends to be legitimate software in order to trick users into installing it and carrying out malicious activities.

1.2.2 CIA Triad

CIA triad is an information security model used to protect data and maintain secure, reliable systems. It guides policies to ensure information remains confidential, accurate and accessible to authorized users.

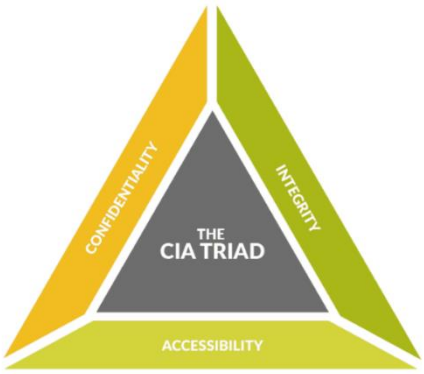


Figure 1.1: CIA Triad

The three core principles are:

i. **Confidentiality:** It is the practice of keeping sensitive data private and blocking access by unauthorized people. Different ways to protect confidentiality are:
   - **Encryption:** It helps maintain confidentiality by converting sensitive data into an unreadable format. Only authorized users with the correct key can decrypt and access the original information.
   - **Authentication:** It helps maintain confidentiality by verifying the identity of a user before allowing access to sensitive information.
   - **Authorization:** It determines what an authenticated user is allowed to access.

ii. **Integrity:** It means ensuring that data and system remain accurate, complete, and trustworthy by protecting them from unauthorized modification, deletion or tempering. Different methods to maintain integrity are:
   - **Hashing:** It helps to maintain integrity by converting data into a fixed-length hash value. If the data is changed, its hash value will also change, indicating that the data has been altered.
   - **Access Control:** It helps maintain integrity by limiting who can modify or delete data. It ensures that only authorized users can make changes, preventing unauthorized alteration.
   - **Digital Signature:** It maintain **i**ntegrity by verifying that data has not been changed or tampered with after it was signed.

iii. **Accessibility:** It means ensuring that authorized users can access systems, applications, and data when they need them**.** Different ways to maintain accessibility are:
   - **Regular Backups: It** helps to maintain accessibility/availability by creating copies of important data that can be restored when the original data is lost, damaged, or encrypted by an attack.
   - **System Monitoring:** It maintains the accessibility by checking the computer system, networks, servers and applications continuously to make sure they are working properly and remain available to authorized users

1.2.3 OSI Model

OSI stands for Open System Interconnection. It is a conceptual seven-layer framework created by the International Organization for Standardization to explain how data is send and receive across a network.

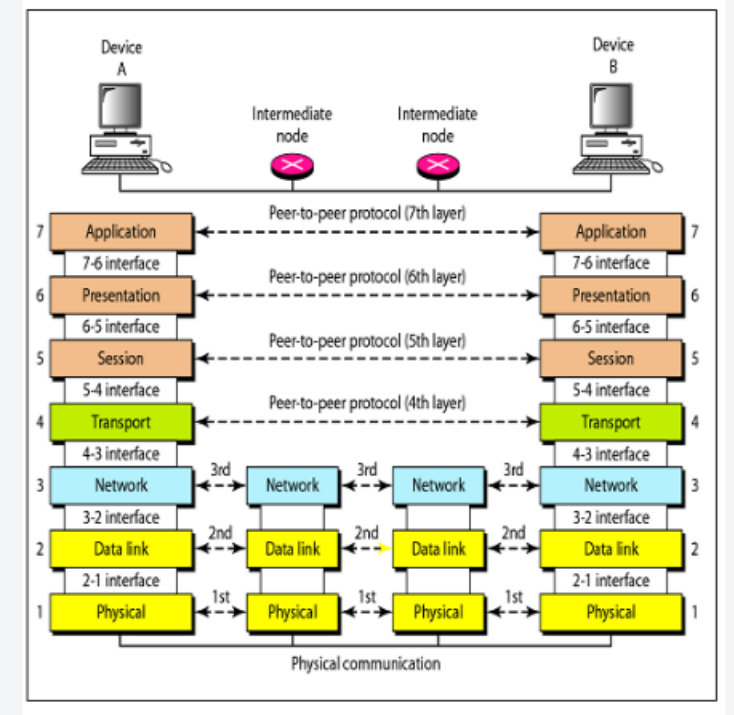


Figure 1.2: OSI Model

i. Physical Layer: The Physical Layer is the first and lowest layer of the OSI model. It is responsible for transmitting raw bits (0s and 1s) through a physical medium. It manages bit synchronization and transmission speed (bit rate) and includes hardware such as cables, hubs, repeaters, and modems. It also supports physical network topologies like Bus, Star, and Mesh and transmission modes such as Simplex, Half-Duplex, and Full-Duplex**.**

ii. Data Link Layer: The Data Link Layer is the second layer of the OSI model**.** It provides reliable node-to-node data delivery by organizing data into frames and using MAC addresses to identify devices. It performs error control, flow control, and access control to ensure safe and efficient data transmission. It is divided into two sublayers:

- **Logical Link Control:** It provides flow control and error control and acts as an interface between the Network Layer and MAC sublayer**.**
- **Media Access Control:** It controls how devices access the transmission medium and uses MAC addresses to identify devices.

iii. Network Layer: The network layer is Layer 3 of the OSI Model, responsible for moving data packets across different networks from a source device to a destination device.

Functions of Network layer are:

- **Logical Addressing:** It is a function of the Network Layer that assigns IP addresses to devices so they can be uniquely identified and data can be delivered from **the** source to the correct destination across different networks.
- **Routing:** It is a function of the Network Layer to manage routes **to** deliver packets, and to determine the best path for data packets to travel from the source device to the destination device across different networks.
- **Path Determination:** It selects the best path or route for data packets to travel from the source to the destination based on factors such as distance, speed and network connections. It uses protocols like: OSPF (Open Source Shortest Path First), BGP(Border Gateway Protocol), IS-IS (Intermediate System to Intermediate System) to determine the best possible path for data delivery.

iv. Transport Layer: Transport layer controls the reliability of communication through:

- **Segmentation:** In Segmentation data received from session layer is divided into small data units called segments. Each segments contains source and destination, sequence number and port number. Port number helps to direct each segment to the correct application and sequence number helps to resemblance segments in the correct order to form correct message at the receiver.
- **Flow Control:** Transport layer controls the amount of data being transmitted.
- **Error Control:** Automatic Repeat Request is used to retransmit the lost or corrupted data. A group of bit checksum is added to each segment by the transport layer to find out the received corrupted segment.

- **Protocols of Transport Layers are:**
  - **Transmission Control Protocol (TCP)**
  - **User Datagram Protocol (UDP)**

v. Session Layer: Session layer helps in setting up and managing connections and labeling of sending and receiving of data followed by terminations connections or sessions. NetBIOS is an example of a service associated with the Session Layer. APIs can provide applications with access to session-related functions.

vi. Presentation Layer: This layer receives the data from application layer. The data is in the form of numbers and characters. Functions of presentation layer are:

- **Translation:** Presentation layer converts this characters and numbers to machine understandable binary format.
  For e.g.: Conversion of ASCII to EBCDIC
- **Data Compression:** It reduces the number of bits needed for transmission, improving network efficiency and speed.
- **Encryption/Decryption:** It secures data by converting plaintext into ciphertext and back, typically handled by protocols like TLS/SSL.

vii. Application Layer: The Application Layer is the topmost OSI layer that provides network services directly to applications and users. The main functions of application layer are:

- Provides network services
- Supports application protocols like HTTPS, FTP, SMTP, DNS, etc.

1.2.4 TCP/IP

TCP/IP (Transmission Control Protocol/Internet Protocol) is a layered networking framework that explains how data is communicated between devices over a network using standardized protocols to ensure reliable and efficient transmission. It has 4 layers:

i. Application Layer
ii. Transport Layer
iii. Internet Layer
iv. Network Layer

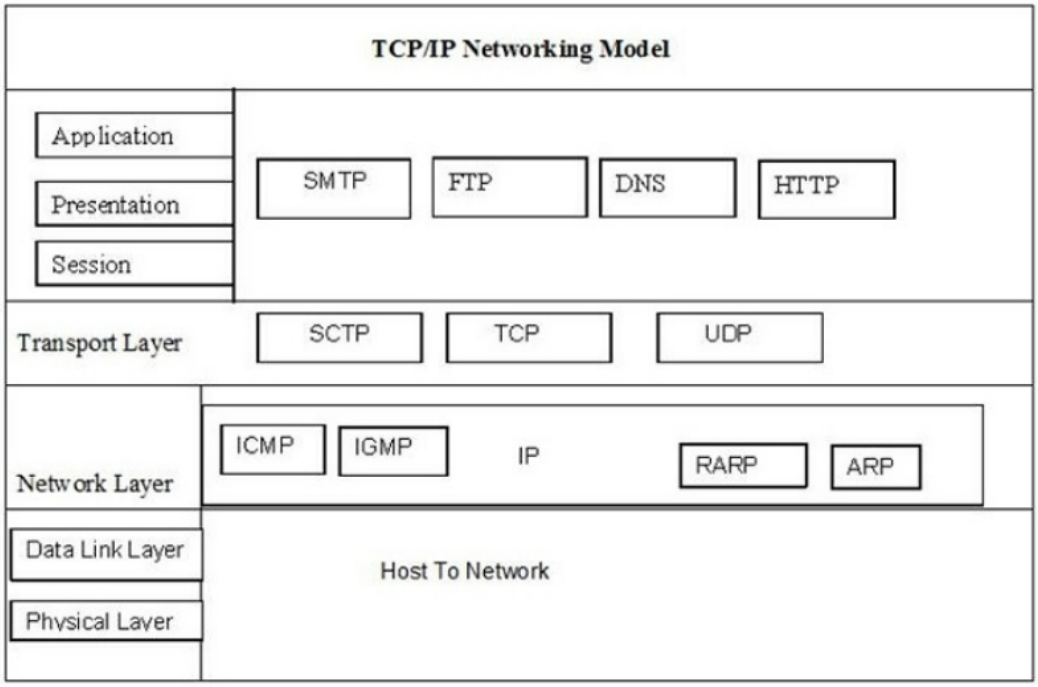


Figure 1.3: TCP/IP Model

i. Application Layer:

- It is the topmost layer in TCP/IP model.
- It is responsible for handling high level protocols such as HTTP, FTP, SMPT, and DNS.
- This layer allows the user to interact with the application and manage sessions to track ongoing connections.
- When application layer protocol wants to communicate with another application layer, it forwards its data to transport layers.

ii. Transport Layer: This layer is responsible for the reliability, flow control and correction of data which is being send over the network. The protocols used in this layer are:

- TCP: checks for errors, resends lost data, and ensures correct order.
- UDP: Provides low-latency, connectionless delivery without error checking

iii. Internet Layer: This layer is responsible for logical addressing, routing and packaging data into packets to send them across networks. Its main functions are:

- **Logical Addressing:** Uses IP addresses to identify source and destination devices.
- **Routing:** Determines the best path for packets to reach the destination across networks.
- **Packet Encapsulation:** Encapsulates transport-layer segments into IP packets/datagrams by adding an IP header.

iv. Network Layer:

- It is the lowest layer of TCP/IP model.
- It defines how the data should be sent physically through the network.
- This layer is responsible for the transmission of the data between two devices on the same network.
- Its function is to encapsulate the IP diagram into frames transmitted by networks and mapping of IP address into physical address.
- The protocol used by this layer are ethernet, token ring, etc.

1.2.5 IP Address

An IP address (Internet Protocol Address) is a unique number assigned to every device connected to a computer network. There are 4 types of IP address:

i. **Public IP Address:** Used to identify a device/network on the Internet.
ii. **Private IP:** Used within a local network, such as a home, school, or office.
iii. **Static IP:** Remains fixed and normally does not change.
iv. **Dynamic IP:** Assigned automatically and may change over time.

- **Versions of IP Address**

i. IPv4 (Internet Protocol Version 4):

- IPv4 uses 32 bits**.**
- It Provides 2^{32} possible addresses.
- It is divided into 4 octets, each containing 8 bits.
- Each octet ranges from 0–255**.**
- Written in dotted-decimal notation**.**
  Example: 192.168.1.2
- It has 3 parts: Network part, Host part, Subnet part.
- Types of IPv4 are: Unicast, Broadcast and multicast

ii. IPv6 (Internet Protocol Version 6):

- IPv6 uses 128-bit addresses
- It Provides 2^{128} possible addresses**.**
- It is written in hexadecimal, separated by colons (:).
- It Contains 8 groups, each of 16 bits.

Classes of IPV4 Address

Table 1.1: Classes of IPv4

IP Class	Address Range	Maximum number of networks
Class A	1-126	$126(2^7-2)$
Class B	128-191	16384
Class C	192-223	2097152
Class D	224-239	Reserve for multitasking
Class E	240-255	Reserved for research

1.2.6 TCP VS UDP

Table 1.2: TCP VS UDP

Feature	TCP	UDP
Full Form	**Transmission Control Protocol**	**User Datagram Protocol**
Definition	TCP is a connection-oriented and reliable transport-layer protocol that ensures data is delivered accurately, completely, and in the correct order.	UDP is a connectionless transport-layer protocol that sends data quickly without guaranteeing delivery, order, or error recovery.
Data Transmission	In TCP, the data is transmitted in a particular sequence which means that packets arrive in-order at the receiver.	There is no sequencing of data in UDP in order to implement ordering it has to be managed by the application layer.
Performance	TCP is slower and less efficient in performance as compared to UDP. Also, TCP is heavy-weight as compared to UDP.	UDP is faster and more efficient than TCP.
Retransmission	Retransmission of data packets is possible in TCP in case packet get lost or need to resend.	Retransmission of packets is not possible in UDP.
Acknowledgement	TCP provides acknowledgement for received packets.	UDP doesn't provides acknowledgement for received packets.

1.2.7 Common ports and Protocols

Ports serve as logical channels that enable communication between devices and applications. Each port is identified by a unique port number ranging from 0 to 65535, assigned by the Internet Assigned Numbers Authority (IANA).

Port Number Categories

i. **Well-Known Ports (0-1023):** Reserved for system services and widely used protocols
ii. **Registered Ports (1024-49151):** Assigned to user processes and specific application upon request.
iii. **Dynamic/Private Ports (49152-65535):** Used by client applications for temporary connections.

Table 1.3: Well-Known Ports

Port No.	**Protocol**
Port 21	FTP (File Transfer Protocol)
Port 22	SSH (Secure Shell)
Port 23	Telnet
Port 25	SMTP (Simple Mail Transfer Protocol)
Port 53	DNS (Domain Name System)
Port 67/68	DHCP (Dynamic Host Configuration Protocol)
Port 80	HTTP (Hypertext Transfer Protocol)
Port 110	POP3 (Post Office Protocol v3)
Port 123	NTP (Network Time Protocol)
Port 143	IMAP
Port 161/162	SNMP
Port 389/636	LDAP/LDAPS
Port 443	HTTPS
Port 445	SMB
Port 993/995	IMAPS/POP3S

Table 1.4: Registered Ports

Ports	Protocols
Port 1080	SOCKS proxy server
Port 1433	Microsoft SQL Server database
Port 1521	Oracle database server
Port 1701	L2TP (Layer 2 Tunneling Protocol)
Port 1723	PPTP (Point-to-Point Tunneling Protocol)
Port 2049	NFS (Network File System)
Port 3128	Squid proxy server
Port 3306	MySQL database server
Port 3389	RDP (Remote Desktop Protocol)
Port 5060/5061	SIP (Session Initiation Protocol)
Port 5222/5223	XMPP (Extensible Messaging and Presence Protocol)
Port 5353	mDNS (Multicast DNS) and Bonjour
Port 5900	PostgreSQL database server
Port 6379	Redis key-value database
Port 6667/6697	IRC (Internet Relay Chat)
Port 8080	HTTP alternative and proxy servers
Port 11211	Memcached distributed caching system
Port 27017	MongoDB database server

Table 1.5: Specialized and Additional Ports

Ports	Protocol
Port 137-139	NetBIOS
Port 500	IKE (Internet Key Exchange)
Port 548	AFP (Apple Filing Protocol)
Port 873	Rsync for remote file synchronization
Port 5355	LLMNR (Link-Local Multicast Name Resolution)
Port 5672	AMQP (Advanced Message Queuing Protocol)
Port 16384-32767	RTP/RTCP (Real-time Transport Protocol)

1.2.8 Linux File System

The Linux file system is the structure used to organize and store files and directories.

The main directory is called the root directory, represented by /.

Table 1.6: Directories

Directory	Purpose
/	Root directory
/home	User's personal files
/etc	Configuration files
/bin	Essential commands/programs
/var	Variable data such as logs
/tmp	Temporary Files
/root	Home directory of the root user
/dev	Device files
/boot	Files needed to boot Linux

Table 1.7: Basic Linux Commands

Command	Purpose
pwd	Show current directory
ls	List files/directories
cd	Change directory
mkdir	Creates directory
touch	Creates a file
cp	Copies files
mv	Move/rename files
rm	Remove files
cat	Display file contents
clear	Clears the terminal
man	Shows command manual

1.2.9 Linux File Permission

Linux permission control who can read, write, or execute a file. There are 3 types:

- r= Read
- w= Write
- x= Execute

And three categories of users:

- u= User/Owner
- g= Group
- o= Others

For e.g.: -rwxr-xr—

Here,

- rwx: Owner can read, write, execute
- r-x: Group can read, execute
- r: Others can read only

- **Common permission Commands are:**

Table 1.8: Linux Permission Commands

chmod	Changes file permissions
chown	Changes file ownership

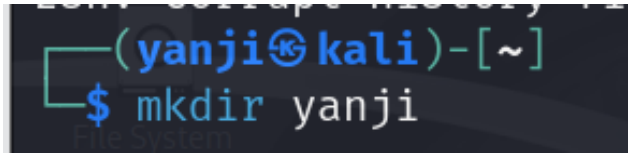
Practical Tasks

1. **Practise and document at least 10 Linux commands.**

i. mkdir

Command: mkdir

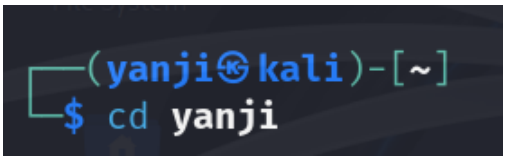
Purpose: Creates a new directory



ii. cd

Command: cd

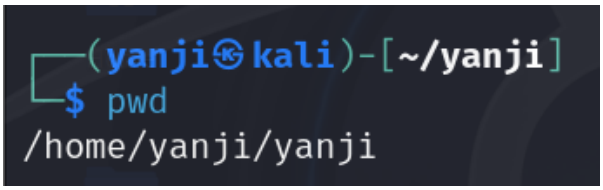
Purpose: Changes to the Documents folder



iii. pwd

Command: pwd

Purpose: Shows the current working directory



iv. touch

Command: touch

Purpose: Creates an empty file



v. echo

Command: echo

Purpose: Displays text or the value of a variable in the terminal

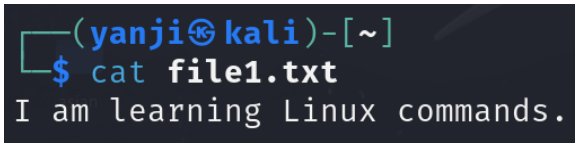


vi. cat

Command: cat

Purpose: Displays the contents of a file

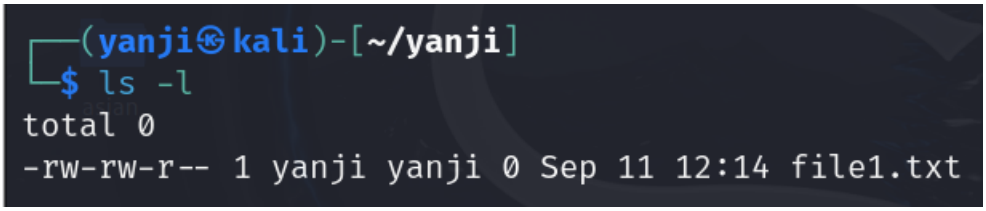




vii. ls -l

Command: ls -l

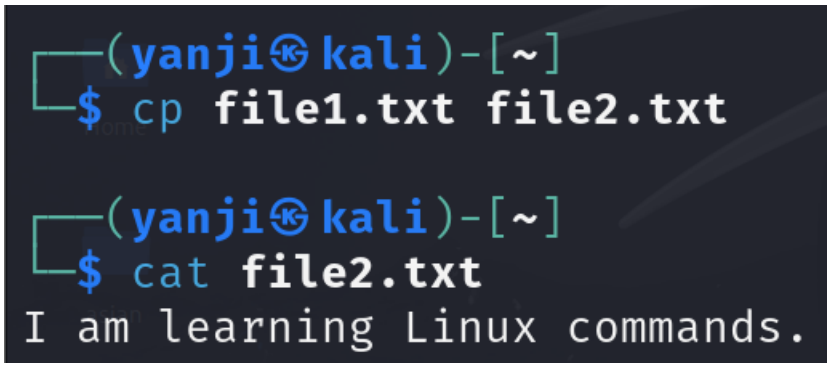
Purpose: lists the files and directories in the current directory in a detailed format.



viii. cp

Command: cp

Purpose: Copies content of one file to another



ix. ls

Command: ls

Purpose: list files and folder



x. rm

Command: rm

Purpose: to remove or delete files in Linux.



2. Create and manage files and directories.

i. **Objective:** To practice creating, viewing, renaming, copying, organizing, and deleting files and directories using core Linux terminal commands.

ii. Procedure:

Step 1: **Create a directory**

The mkdir command creates a new, empty directory.



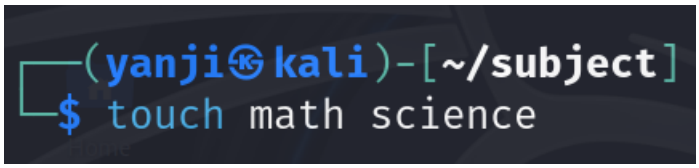
Step 2: Move into the directory

cd changes the current working directory, pwd confirms the current path



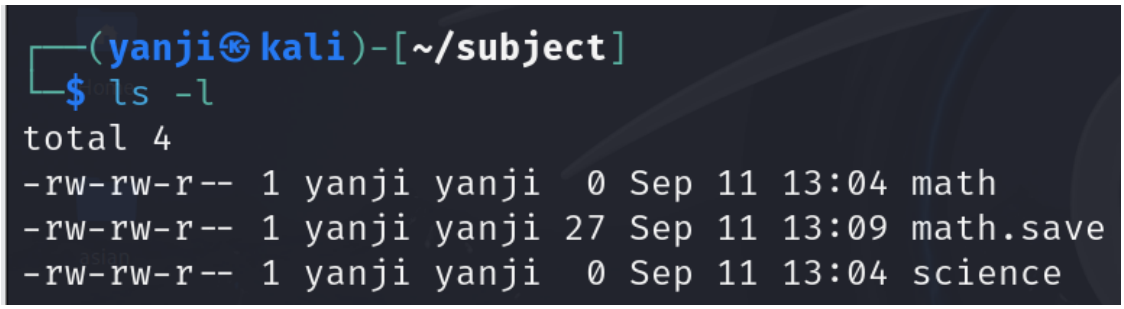
Step 3: Create a file inside it

Touch creates empty files; nano opens a text editor to add real content



Step 4: List and View files

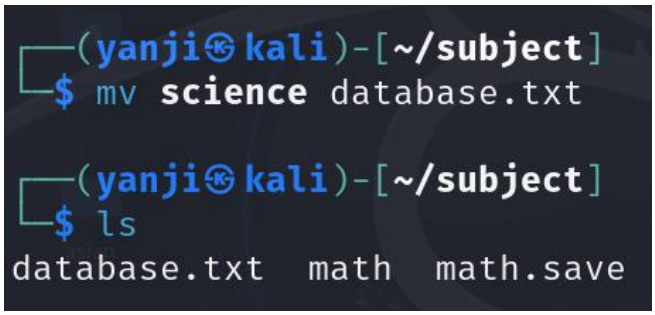
Ls -l shows permissions, owner size, and modified date, cat prints a files content to the terminal.





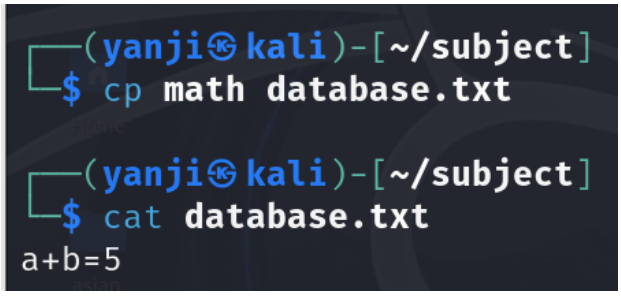
Step 5: Remove or move files

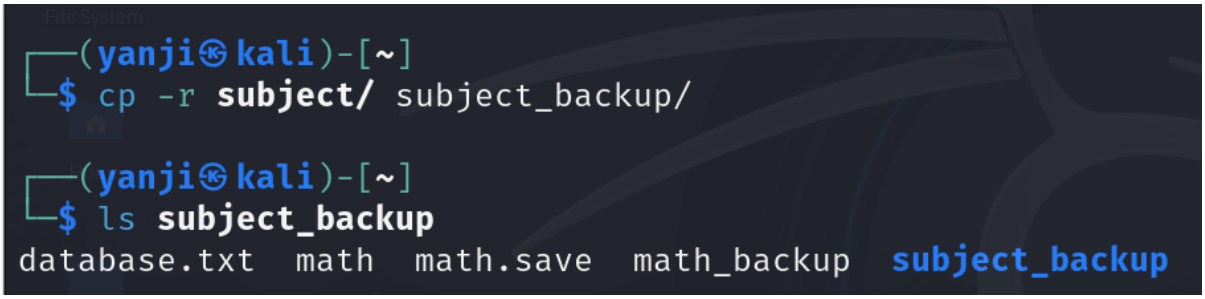
mv renames a file in the same folder, or moves it to a different folder



Step 6: Copy files or folder

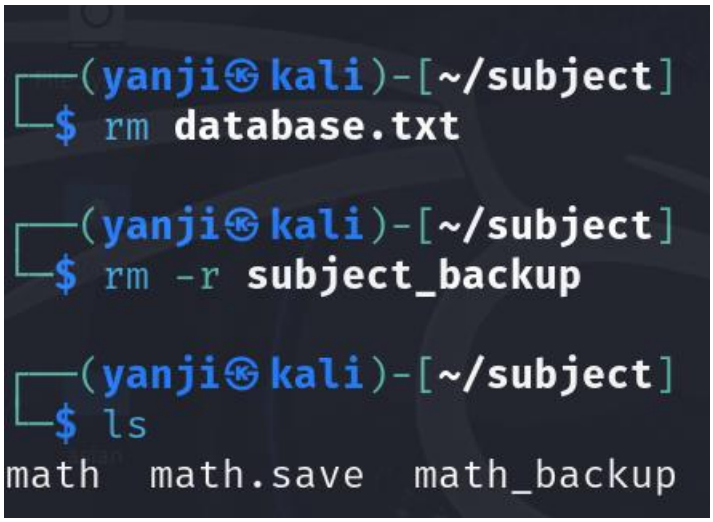
cp duplicates a file content. The -r is required to copy entire folder and its content.





Step 7: Delete files and directories

rm deletes files, rmdir removes only empty directories, rm -r removes a directory and everything inside it.



Conclusion: In this task, I practiced creating, viewing, renaming, copying and deleting files and directories using Linux terminal.

3. Search for files using the terminal

i. Objective: To practice looking files and directories from the terminal using name, type, size, modification time and content-based search techniques.

ii. Procedure

Step 1: Search by exact or partial name

Searches for files/directories by name, starting from a given location. The "." means "search here and in all subfolders."



Step 2: Search for all .txt files



Step 3: Search by File Size



Step 4: Search by Modification Time



4. Prepare table of common ports, protocols and their purposes

Table 2.1: Common Ports, Protocol and Purposes

Port No.	Protocol	Purpose
Port 21	FTP (File Transfer Protocol)	File transfer between client and server
Port 22	SSH (Secure Shell)	Secure encrypted remote login
Port 23	Telnet	Unencrypted remote login (insecure)
Port 25	SMTP (Simple Mail Transfer Protocol)	Sends outgoing email
Port 53	DNS (Domain Name System)	Resolves domain names to IP addresses
Port 67/68	DHCP (Dynamic Host Configuration Protocol)	Auto-assigns IP addresses on a network
Port 80	HTTP (Hypertext Transfer Protocol)	Unencrypted web traffic
Port 110	POP3 (Post Office Protocol v3)	Downloads email from server
Port 123	NTP (Network Time Protocol)	Syncs system clocks
Port 143	IMAP	Manages/syncs email on server
Port 161/162	SNMP	Monitors/manages network devices
Port 389/636	LDAP/LDAPS	Directory service queries (636 = encrypted)
Port 443	HTTPS	Encrypted web traffic
Port 445	SMB	File/printer sharing (mainly Windows)
Port 993/995	IMAPS/POP3S	Encrypted email retrieval

Required Labs

i. Linux Fundamentals Part 1

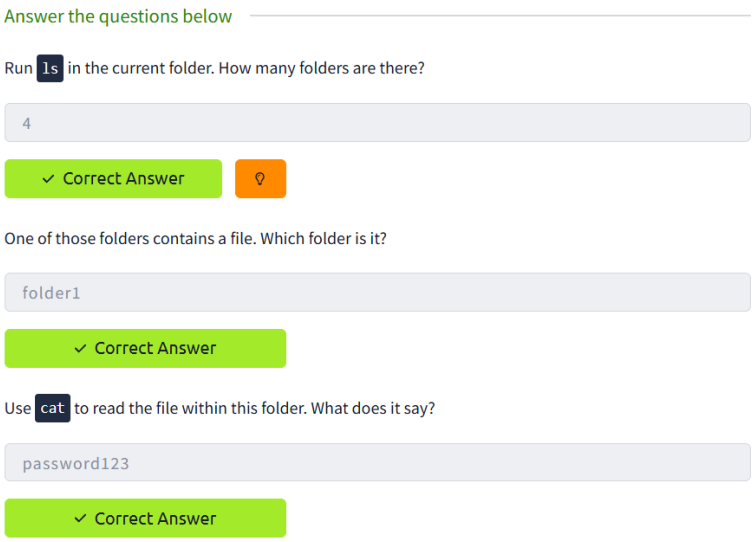


Figure 3.1: Task 1

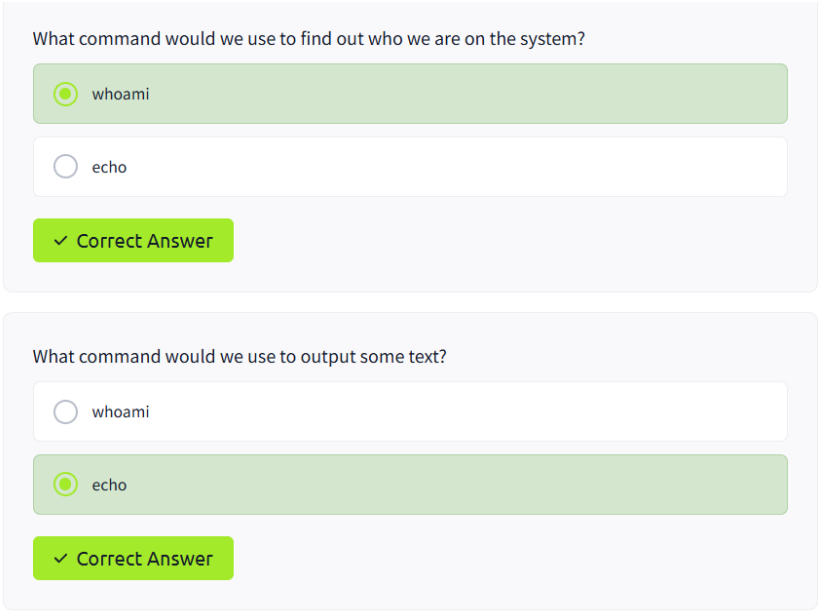


Figure 3.2 : Task 2

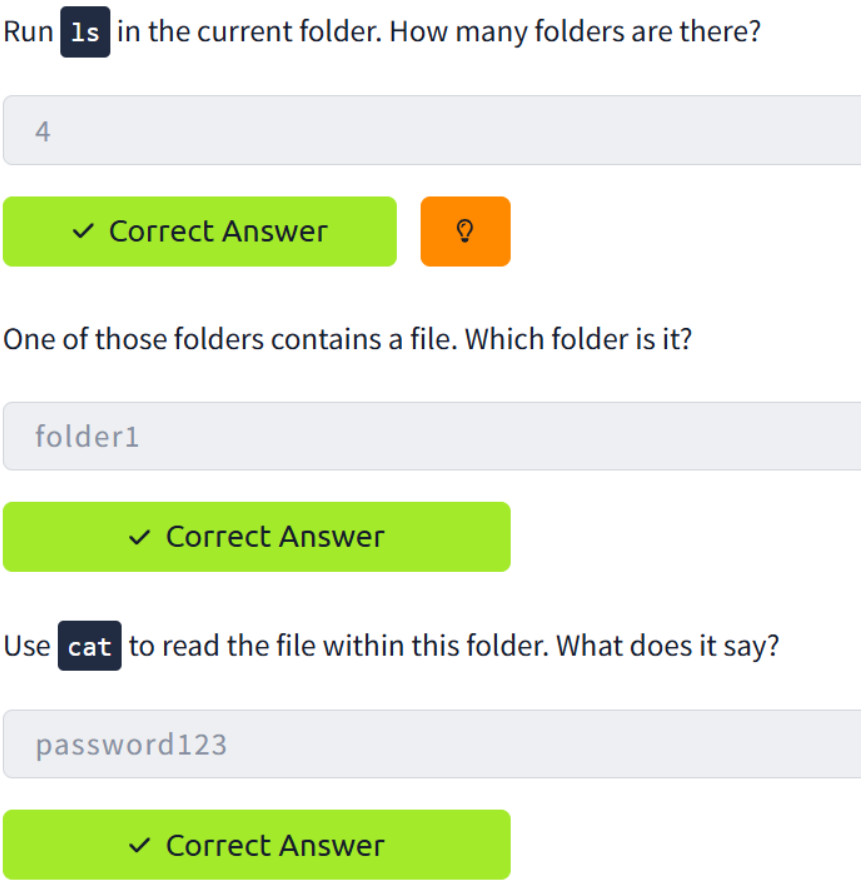


Figure 3.3: Task 3

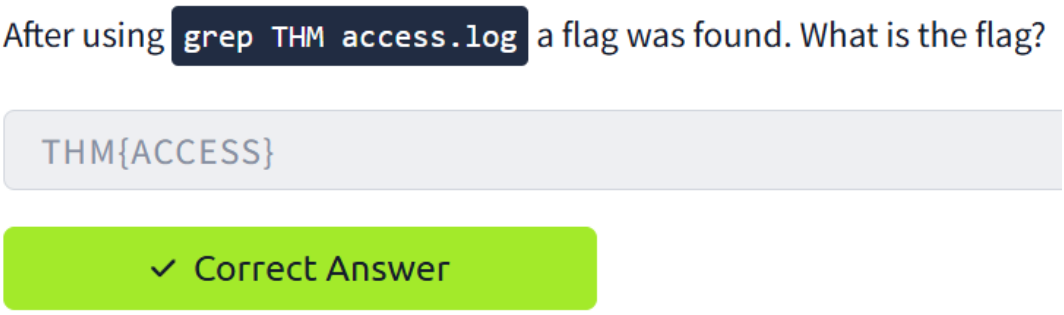


Figure 3.4: Task 4

ii. Networking Concepts

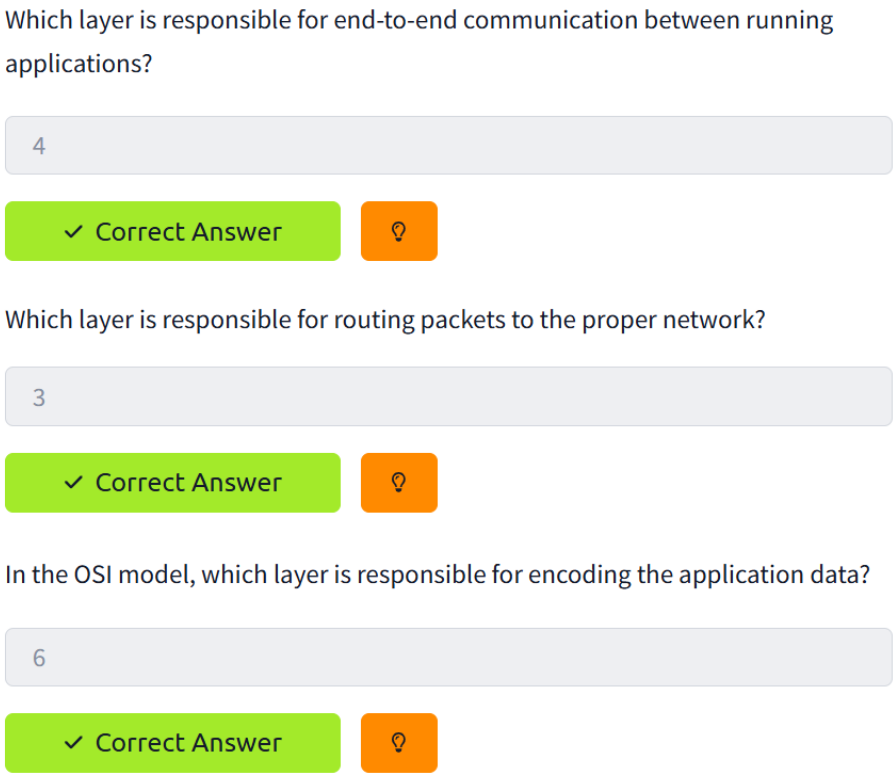


Figure 3.5: Task2

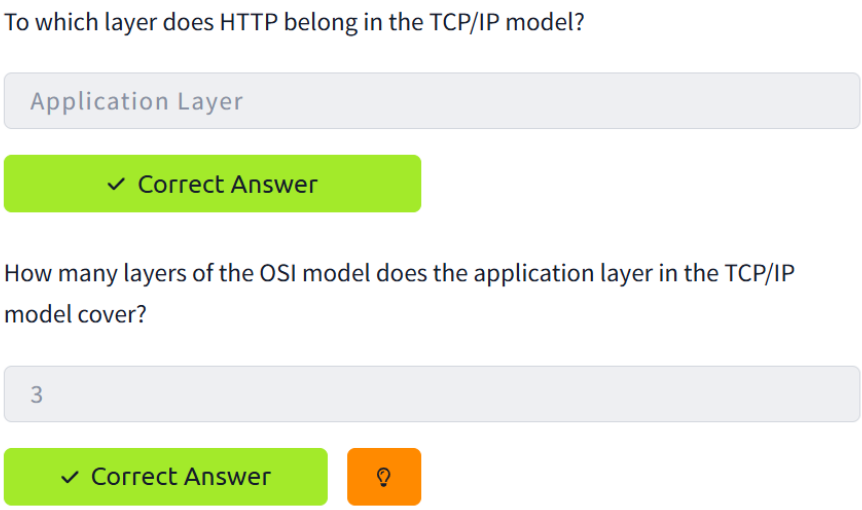


Figure 3.6: Task 3

Which of the following IP addresses is not a private IP address?

- 192.168.250.125
- 10.20.141.132
- 49.69.147.197
- 172.23.182.251

49.69.147.197

✓ Correct Answer

Which of the following IP addresses is not a valid IP address?

- 192.168.250.15
- 192.168.254.17
- 192.168.305.19
- 192.168.199.13

192.168.305.19

✓ Correct Answer

Figure 3.7: Task 4

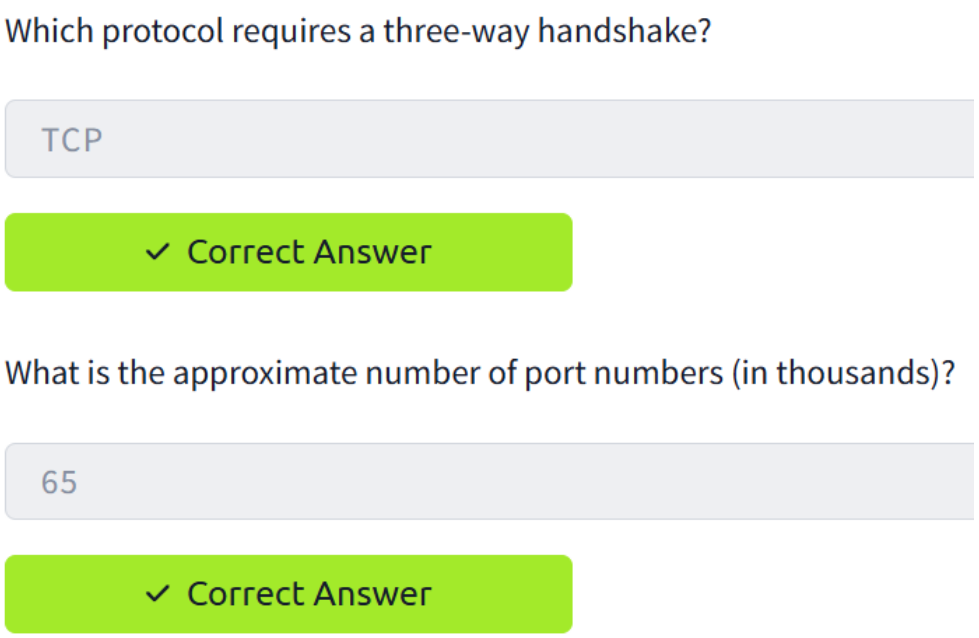


Figure 3.8: Task 5

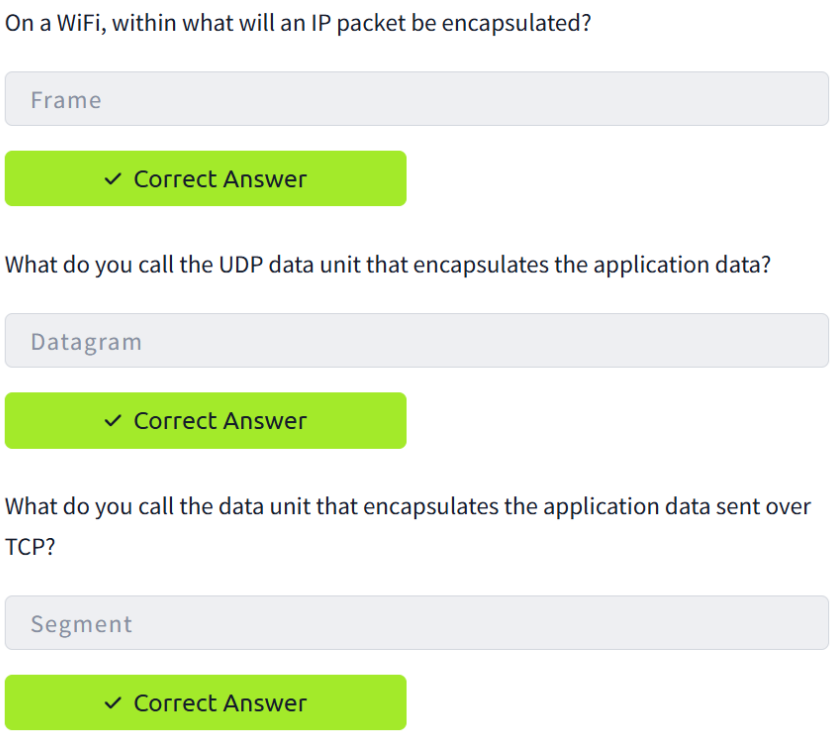


Figure 3.9: Task 6

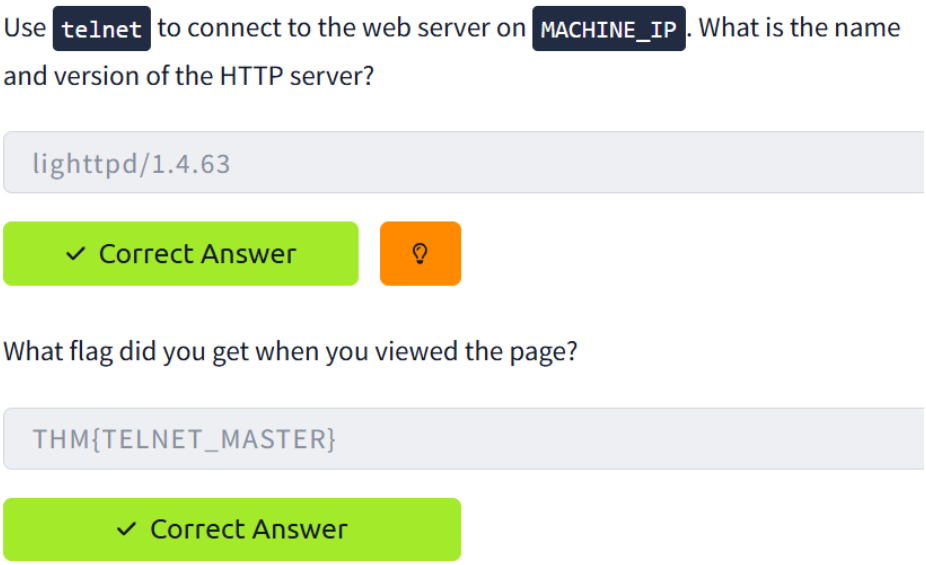


Figure 3.10: Task 7

- **Key Learnings**

I learned the basics of cybersecurity, Linux, and networking. I also got practical experience with Linux commands, file permissions, IP addresses, protocols, and common network ports. The TryHackMe labs helped me understand how these concepts are used in real situations.

- **Challenges Faced**

I found the TryHackMe Networking Concepts lab challenging because some networking terms were new to me. After reviewing the concepts and following the instructions, I was able to understand them better and complete the lab.

- **Conclusion**

Overall, I learned the basics of cybersecurity, the CIA Triad, OSI and TCP/IP models, IP addresses, TCP and UDP, common ports and protocols, HTTP/HTTPS, FTP, SSH, SMB, DNS, and the Linux file system. I also practiced Linux commands, file management, and permissions. The TryHackMe labs helped me understand these topics through practical experience.